OpenGeoSys
OPEN-SOURCE MULTI-PHYSICS

Modelling of flow and transport process in fractured rock using OGS

From Processes in Fracture to Geological Repository Application

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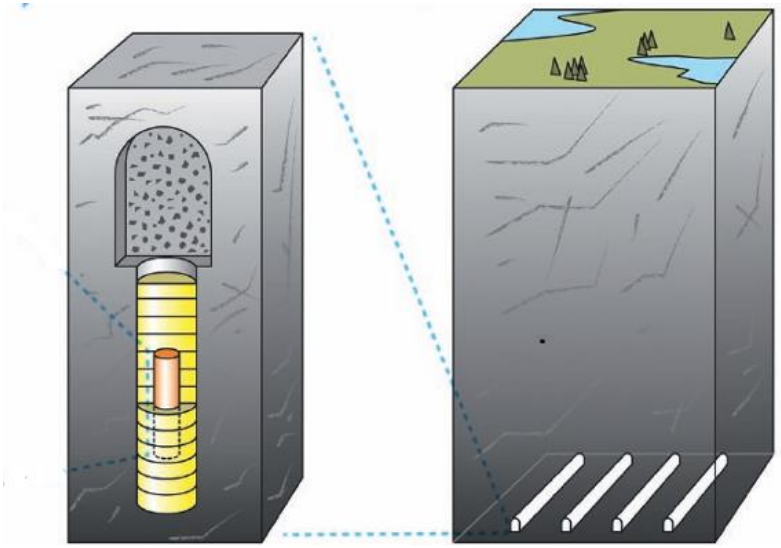
OpenGeoSys Training, Hangzhou 15.10.2026

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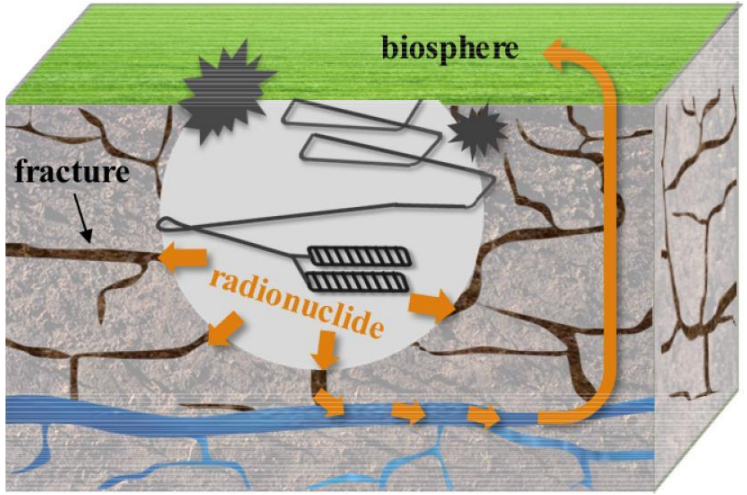
- **Fundamentals of Flow & Transport in Fractured Rock**
- **Modelling Workflow with OpenGeoSys (OGS)**
- **Explicit Single-Fracture Modelling**
- **Real-World Application: 3D Repository Model**
- **Advanced Topic: Stochastic Safety Assessment**

Motivation

- **Cystalline rock** as worldwide potential host rock in deep geological repository for radionuclides
  - China, Finland, Sweden, Canada, Czech Republic, etc
- Fractures are critical for repository performance:
  - **Preferential flow & transport** along connected fractures
  - **Strong heterogeneity and scale effects**
  - **Coupled processes:** stress / temperature → aperture → permeability
  - **Large uncertainty** in fracture geometry, connectivity and propertie



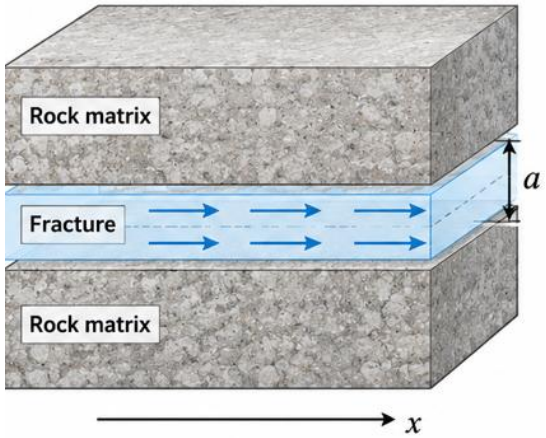
KBS-3 Repository Concept



DOI: 10.32390/ksmer.2023.60.2.112

Flow in fracture

- For laminar flow between parallel plates: $q_f = -\frac{a^3}{12\mu}\left(\frac{\partial p}{\partial x} - \rho \mathbf{g}\right)$



q_f :	volumetric flow rate per unit fracture width	$[\mathrm{m^2\,s^{-1}}]$
a :	fracture aperture	$[\mathrm{m}]$
μ :	dynamic viscosity of the fluid	$[\mathrm{Pa\,s}]$
p :	fluid pressure	$[\mathrm{Pa}]$
x :	coordinate along the fracture	$[\mathrm{m}]$
$\frac{\partial p}{\partial x}$:	pressure gradient along the fracture	$[\mathrm{Pa\,m^{-1}}]$
ρ :	fluid density	$[\mathrm{kg\,m^{-3}}]$
g :	gravitational acceleration	$[\mathrm{m\,s^{-2}}]$

- Equivalent Darcy representation

$$\mathbf{q} = -\frac{k}{\mu}\left(\nabla p - \rho \mathbf{g}\right)$$

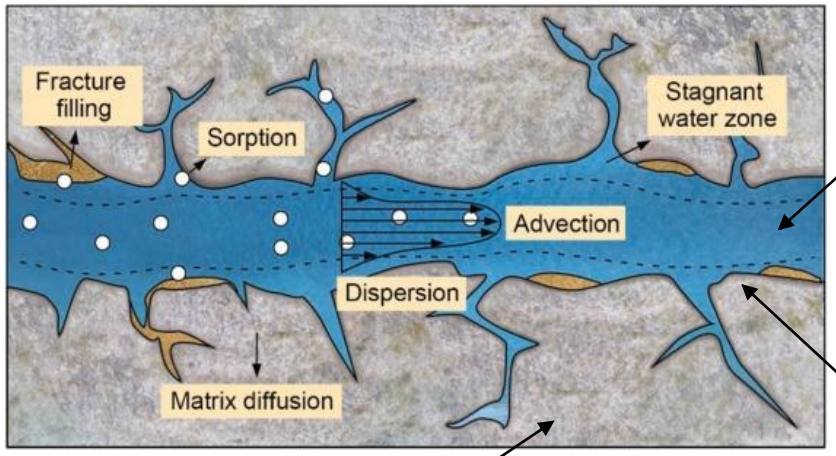
- Cubic law:

$$\boxed{k_f = \frac{a^2}{12}}$$

- Transmissivity

$$T_f = k_f a = \frac{a^3}{12}$$

Transport in fracture



(Zhang et al., 2022)

- **Fracture: Advection, dispersion, matrx diffusion, decay**

$$R_f \frac{\partial c_f}{\partial t} + v\frac{\partial c_f}{\partial z} - D_f \frac{\partial^2 c_f}{\partial z^2} + R_f \lambda c_f = -\frac{q_m}{b}$$

$\underbrace{v\frac{\partial c}{\partial z}}_{\text{advection}}$ $\underbrace{D_f \frac{\partial^2 c}{\partial z^2}}_{\text{dispersion}}$ $\underbrace{q_m}_{\text{matrix diffusion}}$ $\underbrace{\lambda c}_{\text{decay}}$

$$D_f = \alpha_L v + D^*$$

- **Fracture** ⟺ **Matrix: Matrix diffusion**

$$c_m(x = b) = c_f$$

$$q_m = -\theta D_m \left.\frac{\partial c_m}{\partial x}\right|_{x=b}.$$

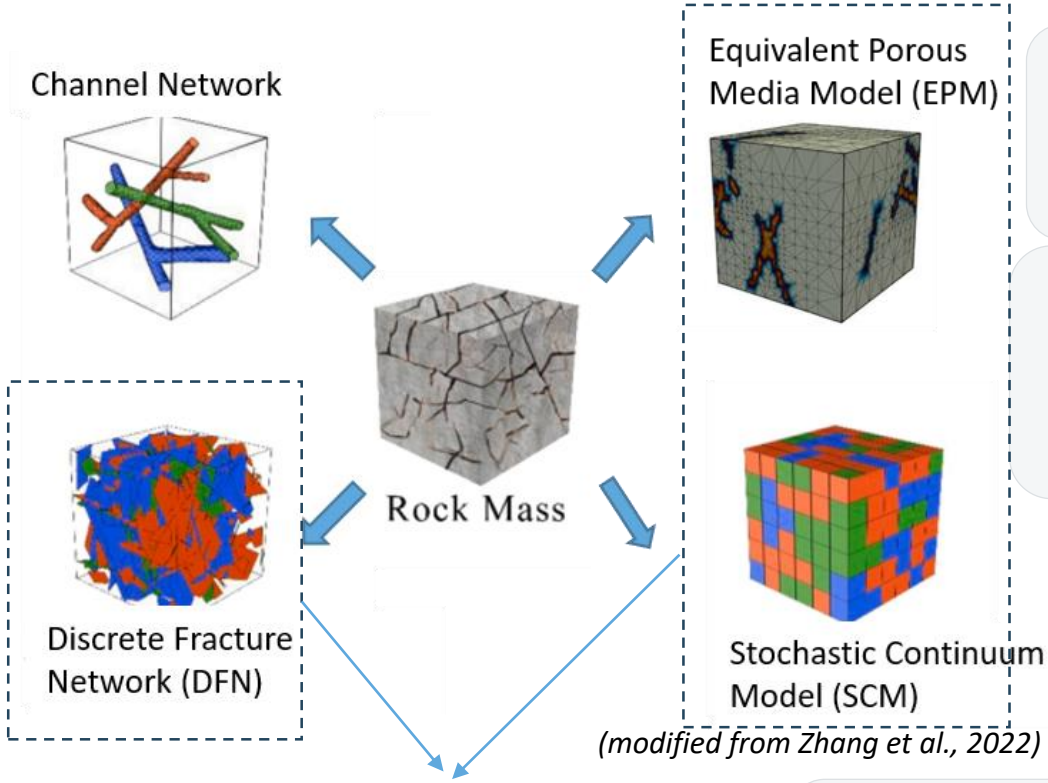
- **Matrix: diffusion +** potentially advection/dispersion

$$R_m \frac{\partial c_m}{\partial t} = D_m \frac{\partial^2 c_m}{\partial x^2} - R_m \lambda c_m$$

$$D_m = \tau D^*$$

Tipps: Fracture–matrix interface requires sufficient spatial resolution

Numerical Representation of Fractured Rock



(modified from Zhang et al., 2022)

Channel Network

Preferential pathways represented as interconnected 1D channels

+ **Efficient for pathway / transport studies**
– Strong geometrical simplification

DFN/DFM

- Individual fractures generated explicitly from statistical geometry and properties
- Selected fractures explicitly embedded in the rock matrix, often as lower-dimensional elements

+ **Best representation of network connectivity**
– High data demand and computational cost

EPM

homogenizes the fracture network, represent the fracture effects through equvalent continuum fields

+ **Simple model setup and conventional continuum mesh, Low computational cost**
– Individual pathways are not explicitly resolved

SCM

Fracture effects represented through spatially varying continuum properties

+ **Efficient at repository / regional scale and for UQ and safety assessment**
– Individual pathways are not explicitly resolved

Repository modelling: DFN → DFM → SCM / EPM

site characterization & connectivity → process-scale fracture–matrix coupling → repository/regional scale & uncertainty

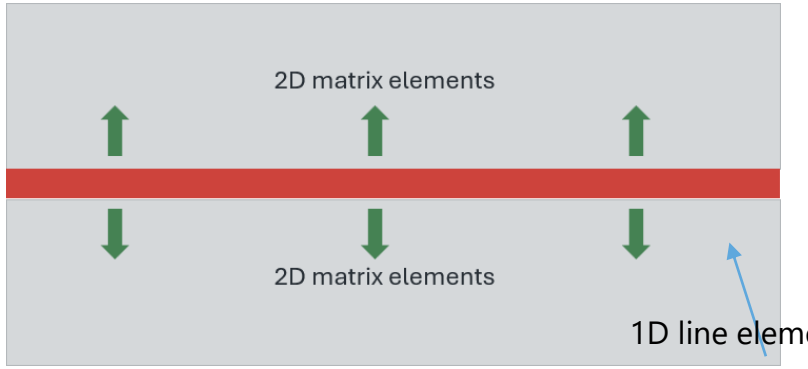
Lower-dimensional interface element (LIE) approach in OGS

https://www.opengeosys.org/6.5.9/docs/benchmarks/hydro-mechanics/lie-hm-linear-single-fracture/

https://www.opengeosys.org/stable/docs/tools/fileio/gmsh2ogs/

1. **Define fracture geometry** — line / surface
2. **Generate computational mesh** — matrix + lower-dim. fracture elements
3. **Identify fracture domains** — MaterialID
4. **Configure OGS project (.prj)** — HM_WITH_LIE fracture properties
5. **Solve coupled fracture–matrix model** — p • u • aperture • flow

Numerical idea



Key OGS concepts / .prj keywords

```
<type>HYDRO_MECHANICS_WITH_LIE</type>

<process_variable>pressure</process_variable>
<process_variable>displacement</process_variable
>
<process_variable>displacement_jump1</process_va
riable>

<fracture_properties>
  <material_id>1</material_id>
  <initial_aperture>aperture0</initial_aperture>
</fracture_properties>

<fracture_model>
  <type>LinearElasticIsotropic</type>
</fracture_model>
```

MaterialID
links .prj fracture properties to lower-dimensional fracture elements

initial_aperture
initial fracture aperture parameter

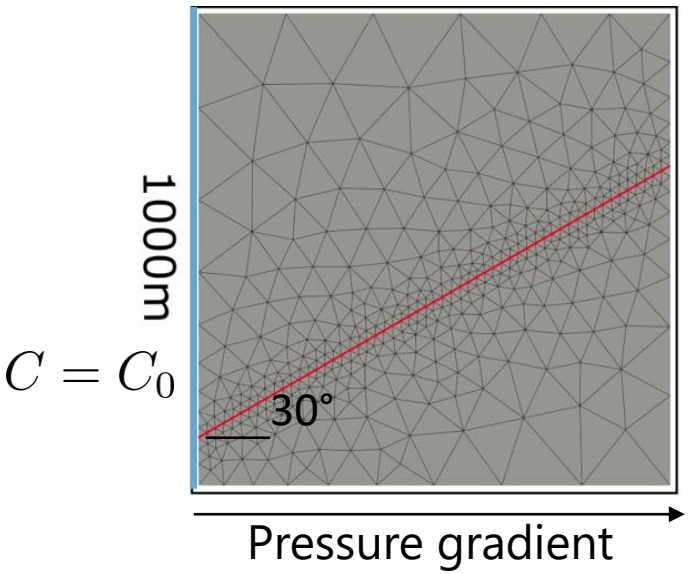
displacement_jump
additional fracture DOF for relative displacement

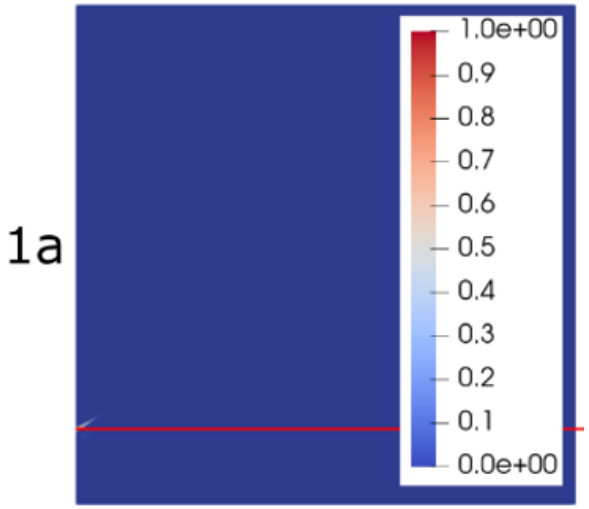
fracture_model
constitutive behaviour, e.g. linear elastic / Coulomb / cohesive

https://www.opengeosys.org/6.5.9/docs/benchmarks/liquid-flow/roughfracture_benchmark/

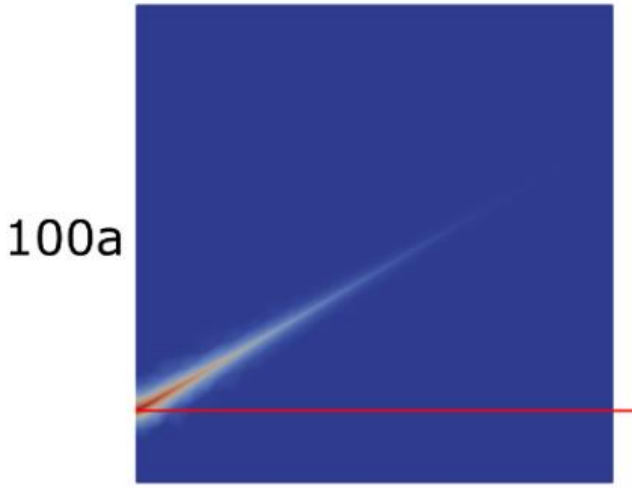
Model verification vs. **Model validation**?

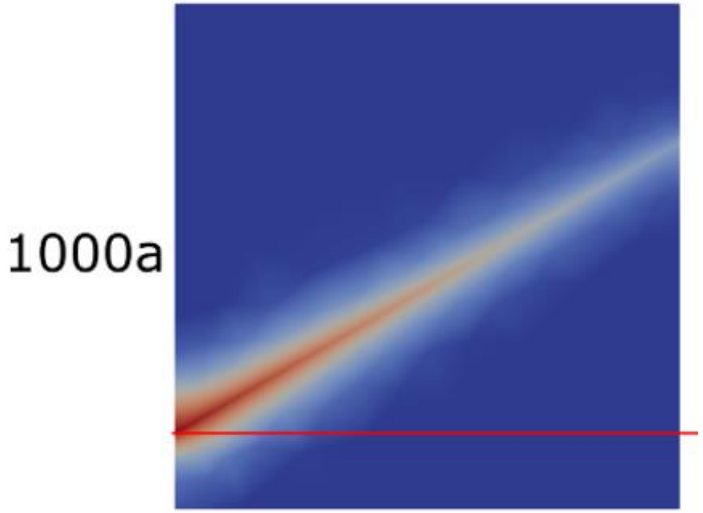
OGS for DFM model

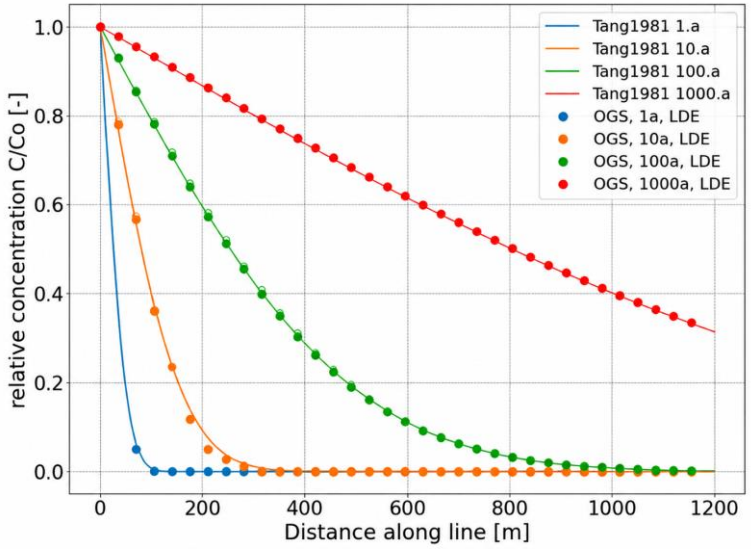










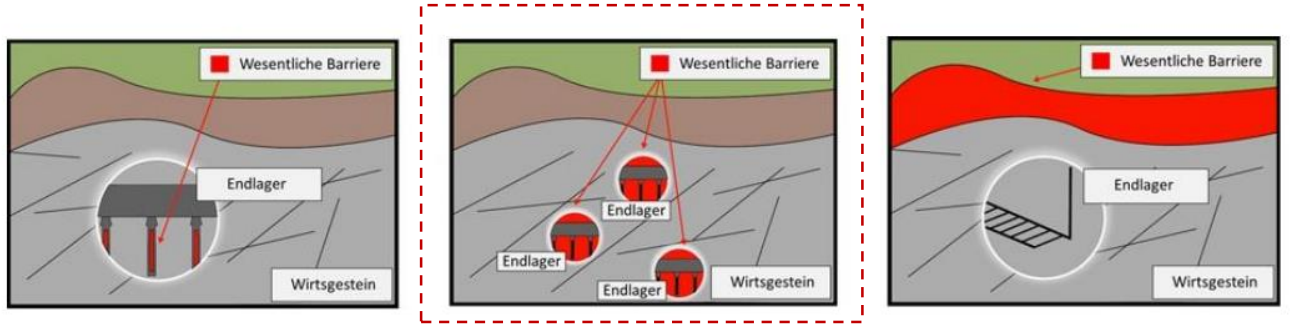


Application case study in DGR

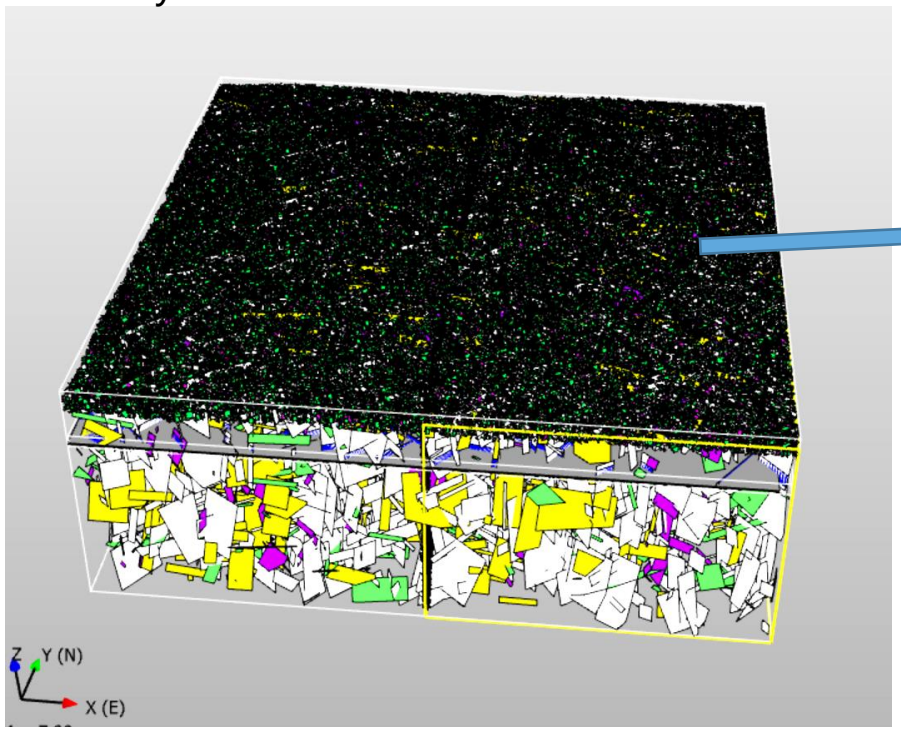
Repository Concepts for Crystalline Host Rock in Germany

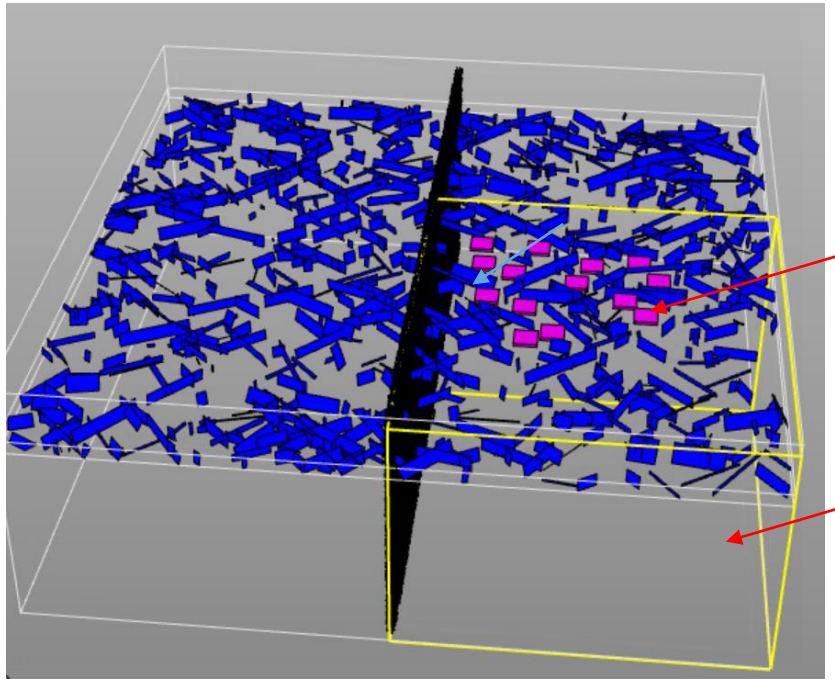
- Christa Project III

I: mKBS-3 *II: m-CPZ* *III: o-CPZ*



Synthetic Fractures





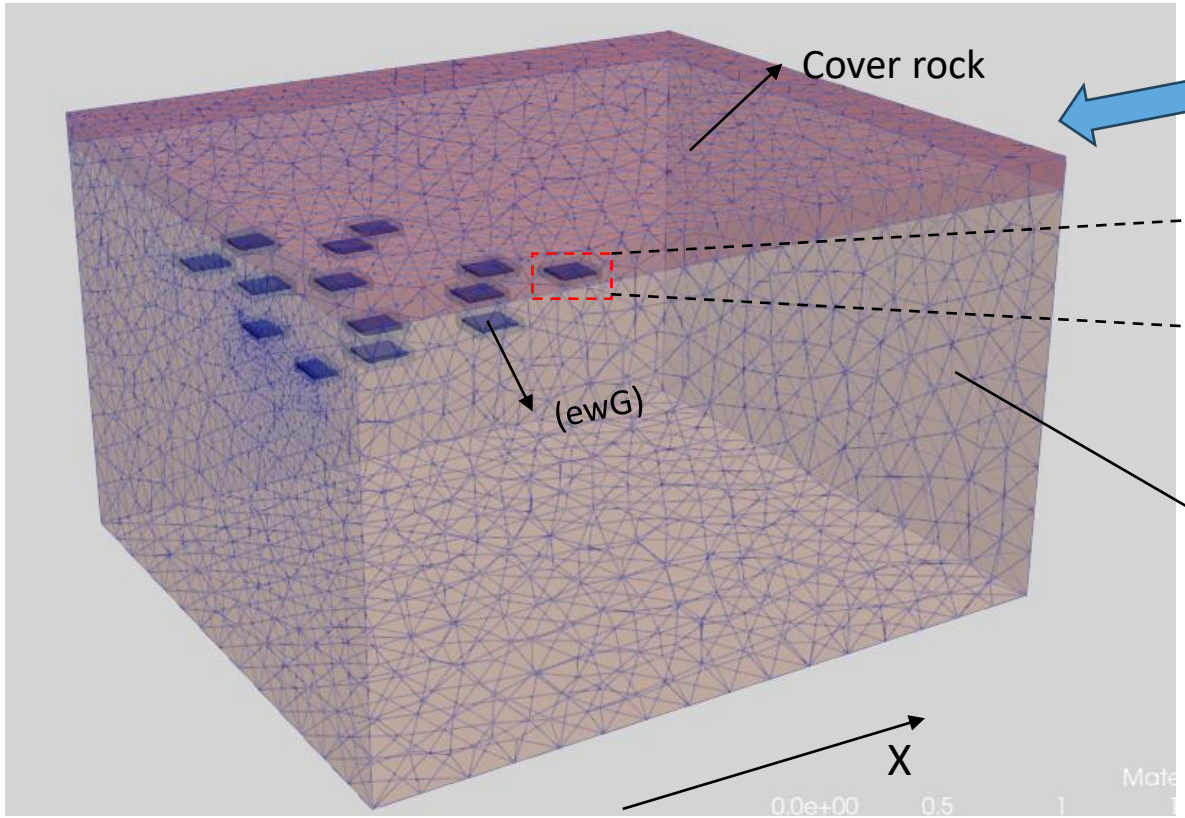
CPZ
Containmination **P**roviding **Z**one

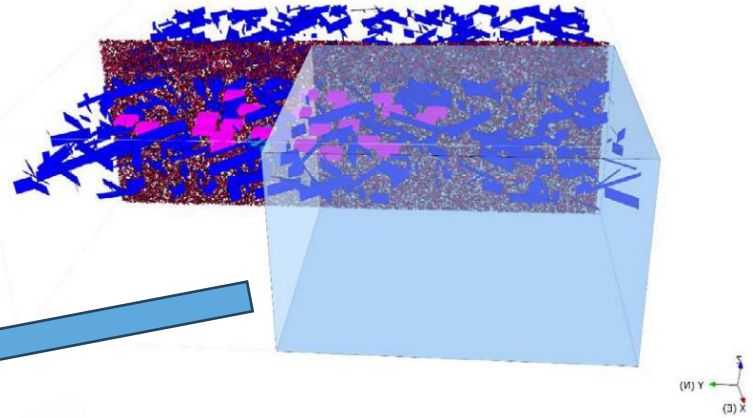
Computational domain

Modelling Concept

- Migration of **Cs-135** and **I-129** over 1 Mio years
  - ***I-129*** *vs.* ***Cs-135****: weak sorption effect, longer half-life time*
- Initial concentration of 0.2 mg/l in repository area
- Hydraulic gradient in x-direction

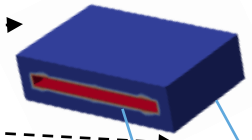
$$\Delta h = 0.2\%$$





CPZ

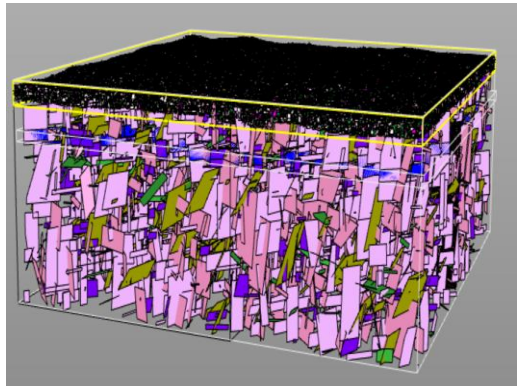
(Christa II-Bericht, BGR 2021)



- **MG1**: Emplacement area
- **MG2**: Geotechnical Barrier (Bentonite)
- **MG3**: Fractured rock

EPM: Workflow for Upscaling and Mapping onto Continnum

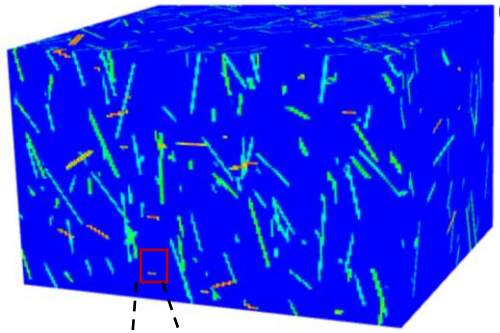
Synthetic fractures generated by FracMan



FracMan Grid

I. Upscaling kf: ODA approach

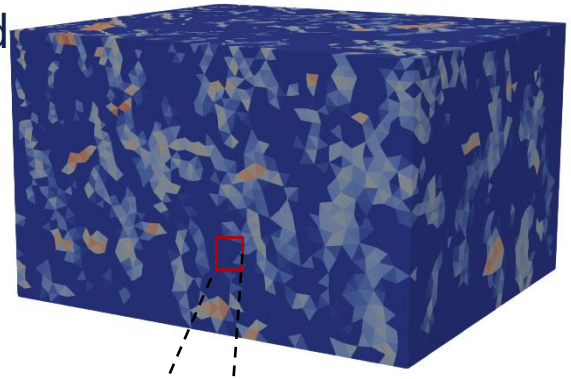
Using FracMan



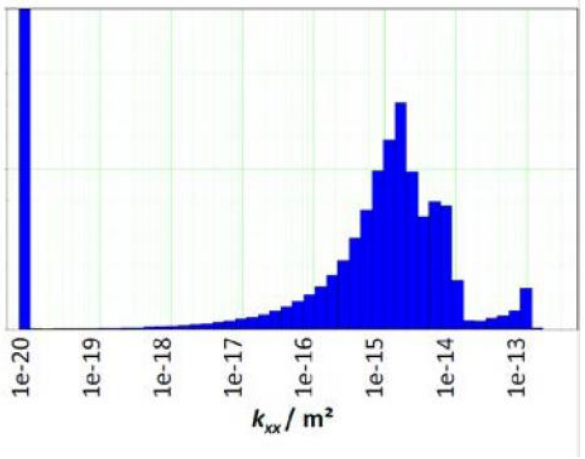
OGS Grid .vtu

II. Mapping on the FE mesh, emble CPZ buffer and

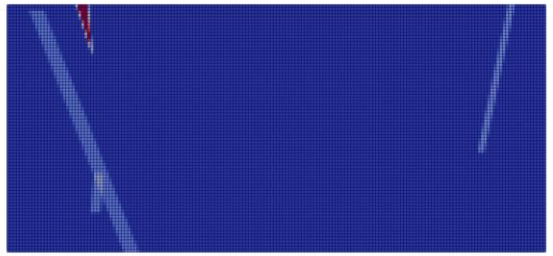
Python script: Geometric mean



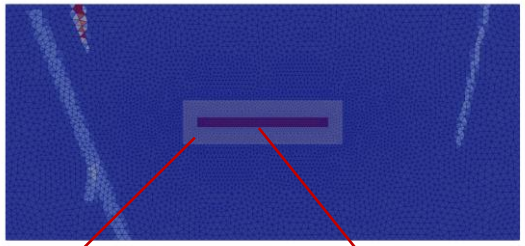
MG3: Permeability distribution in cystalline rock



FracMan



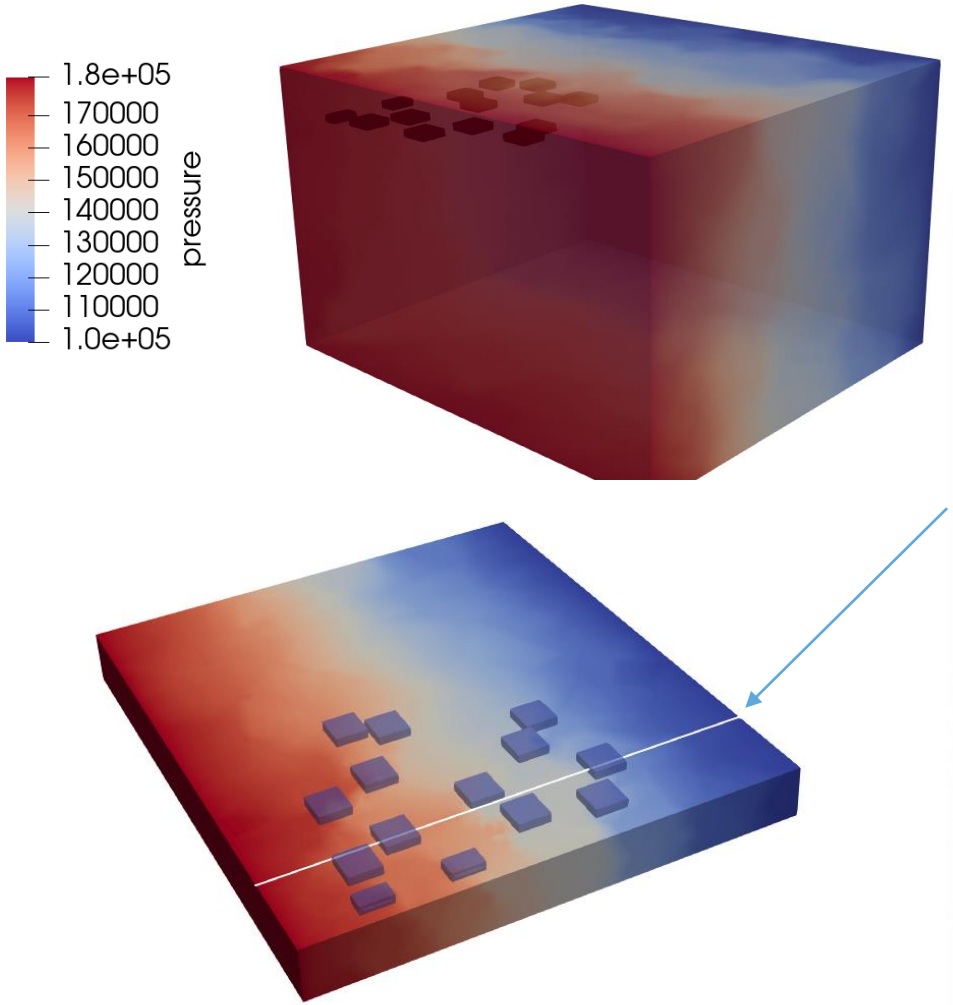
OGS Grid

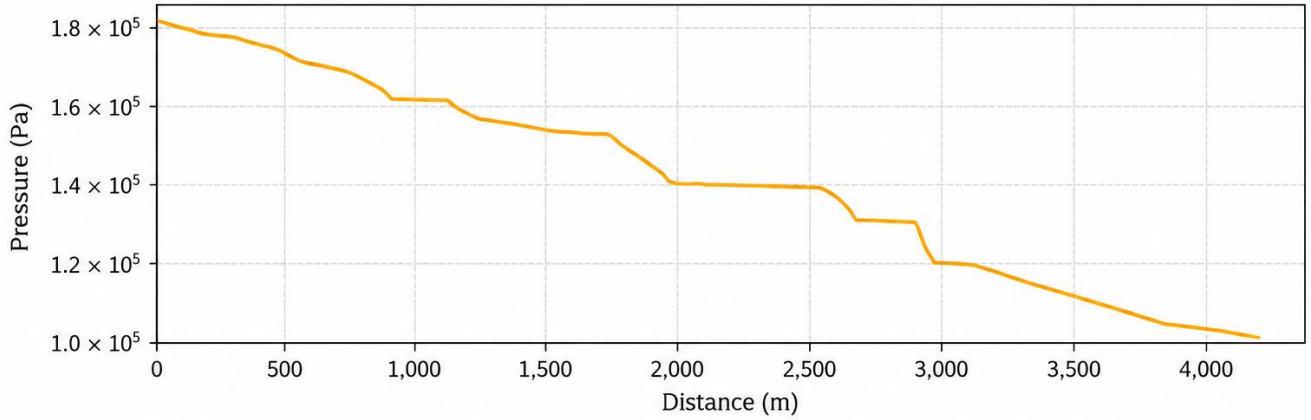


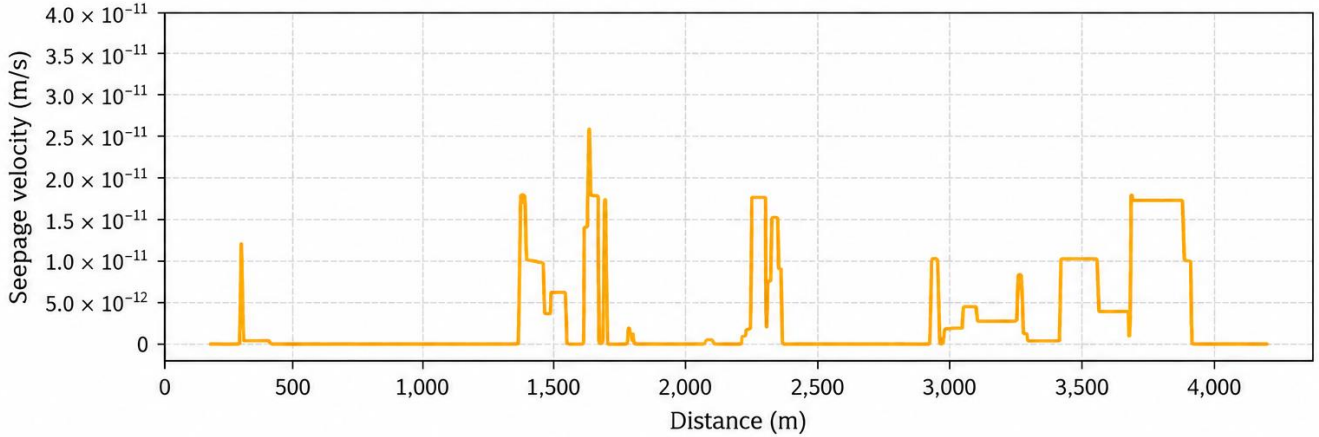
MG1: CPZ, k=1e-20 m^2

MG2: Emplacement area, k= 5e-18 m^2

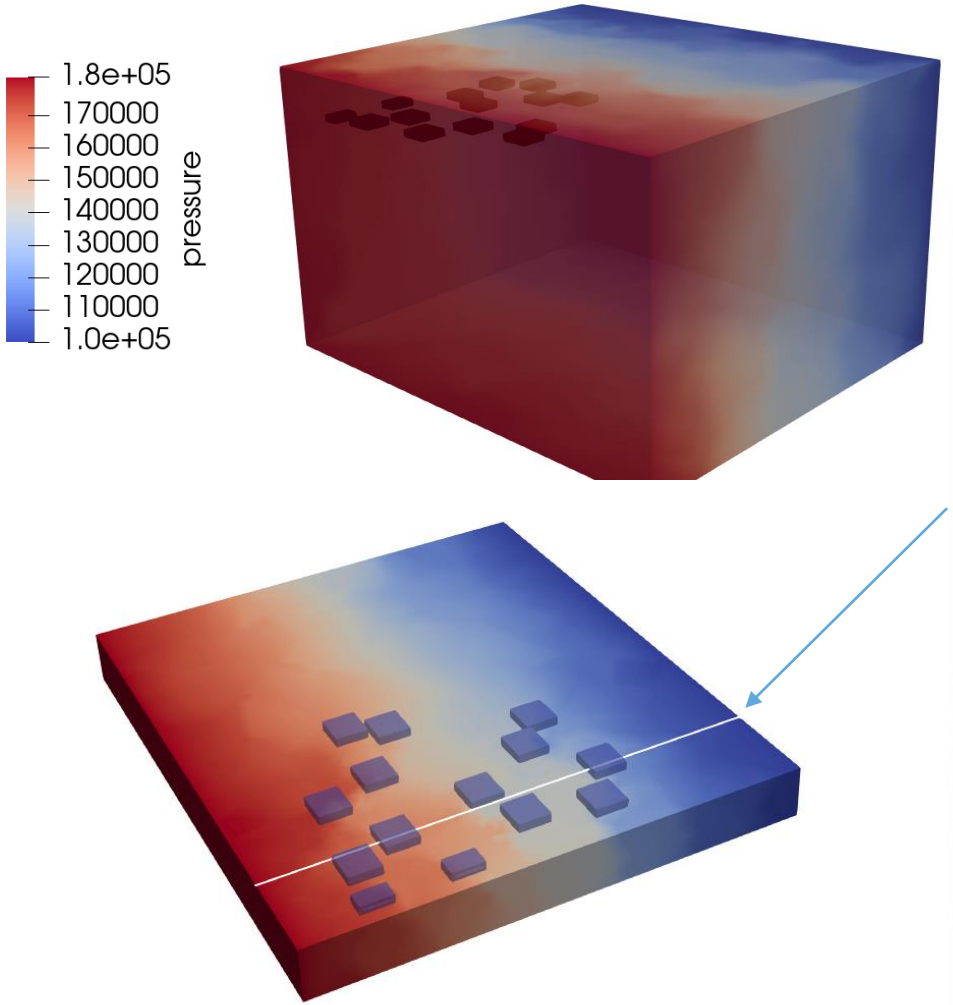
Modelling results: Pressure and Velocity



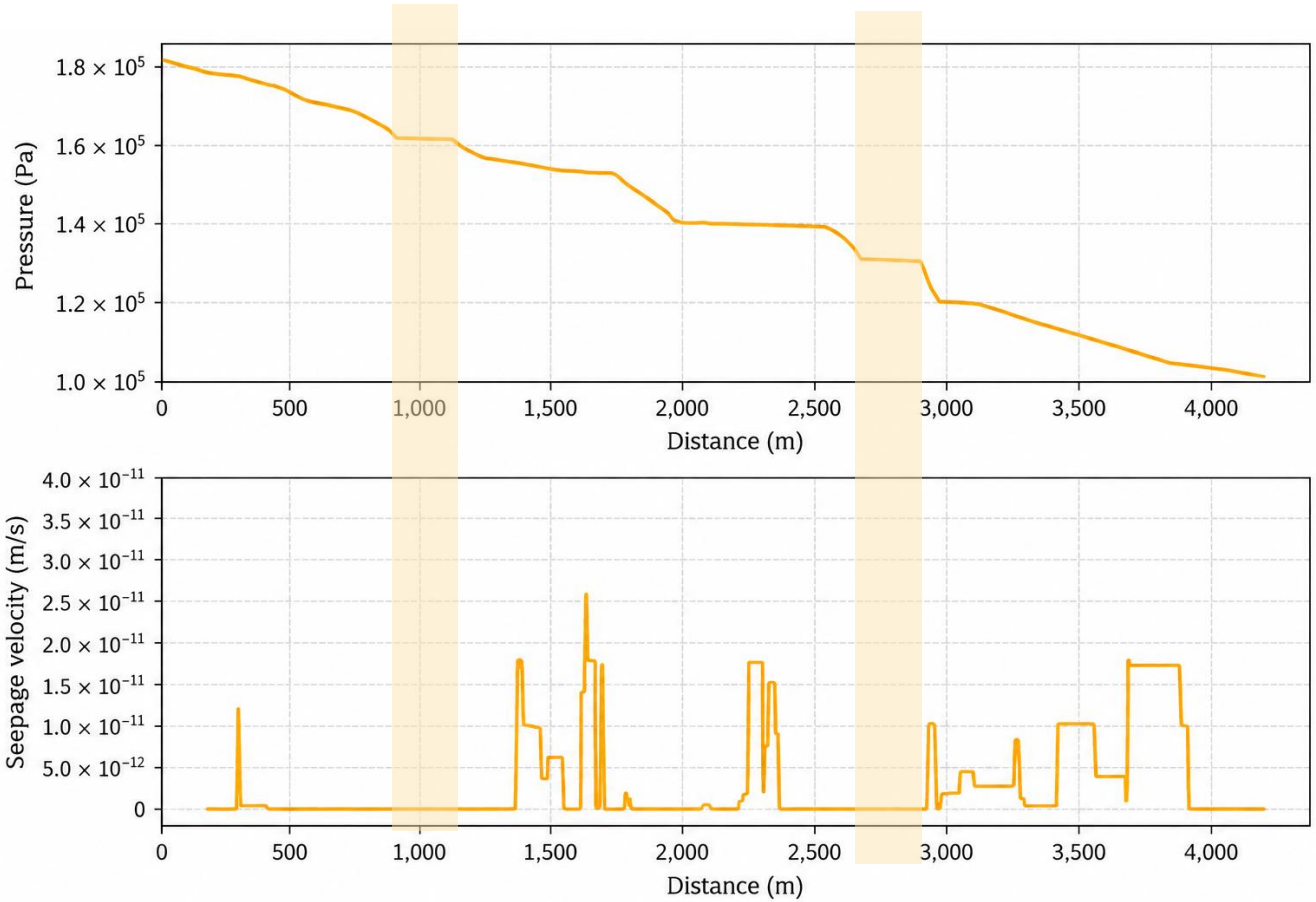




Modelling results: Pressure and Velocity



CPZs

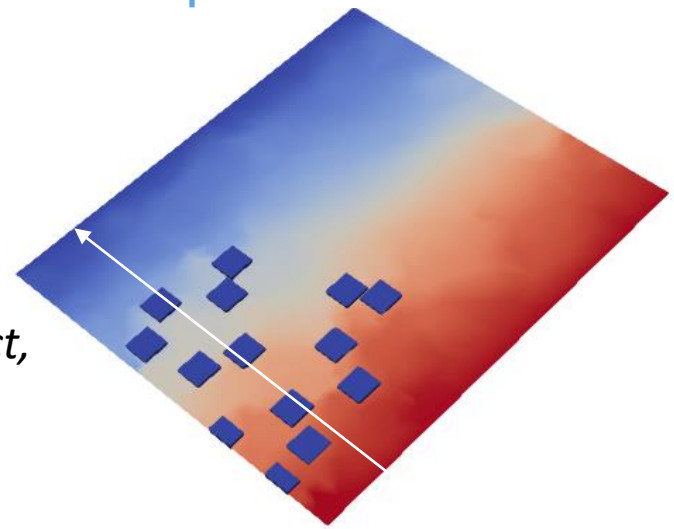


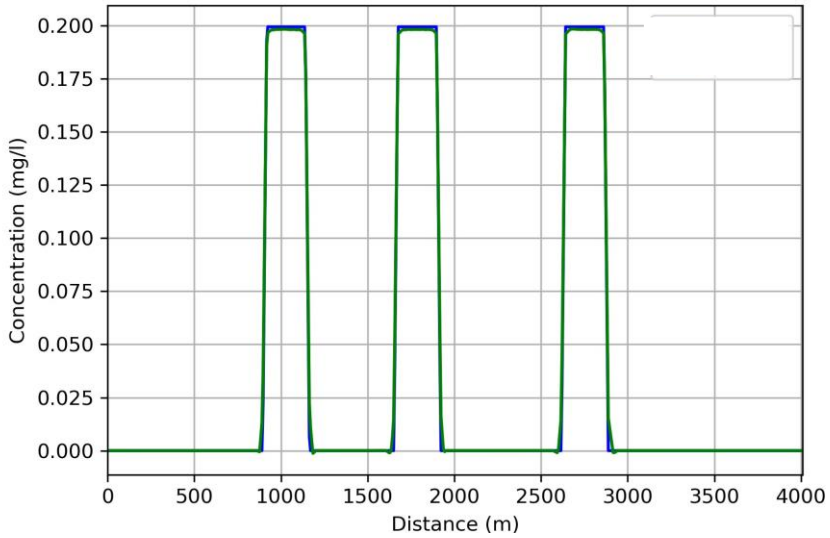
Modelling results: Transport

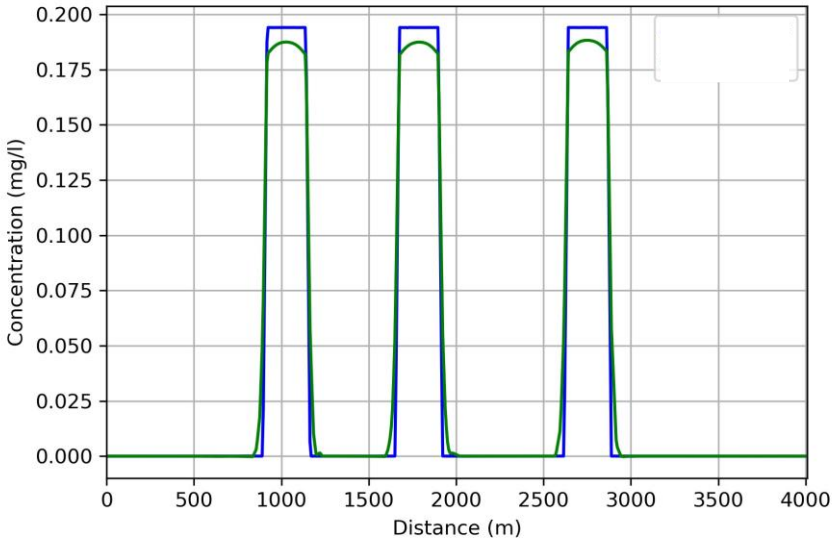
- Concentration profils of **Cs-135** and **I-129:**

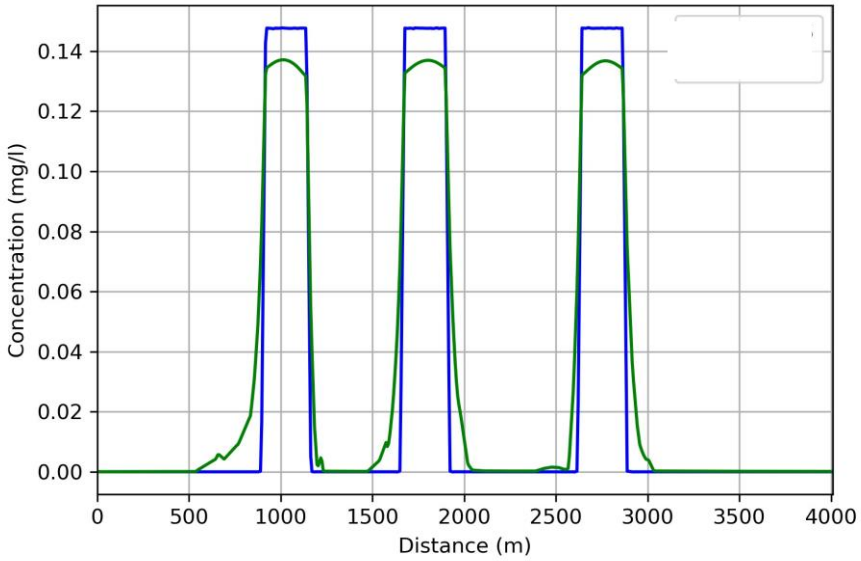
after **10,000**, **100,000** and 1 Mio years

***(I-129** vs. **Cs-135**: weak sorption effect, longer half-life time)*

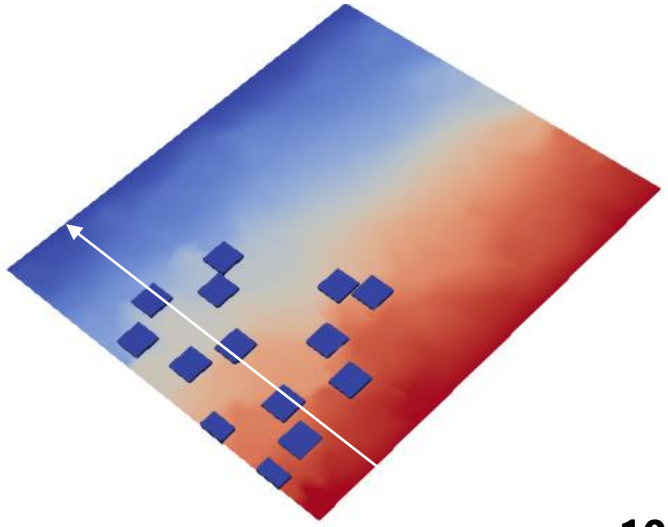




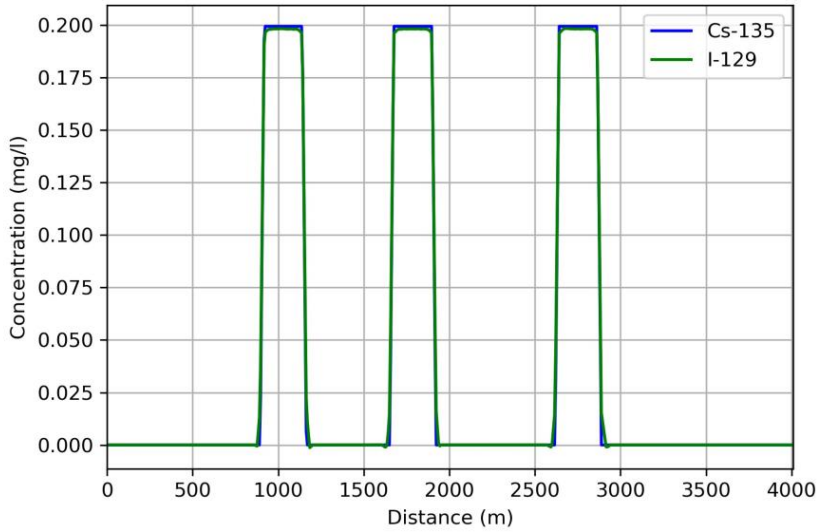




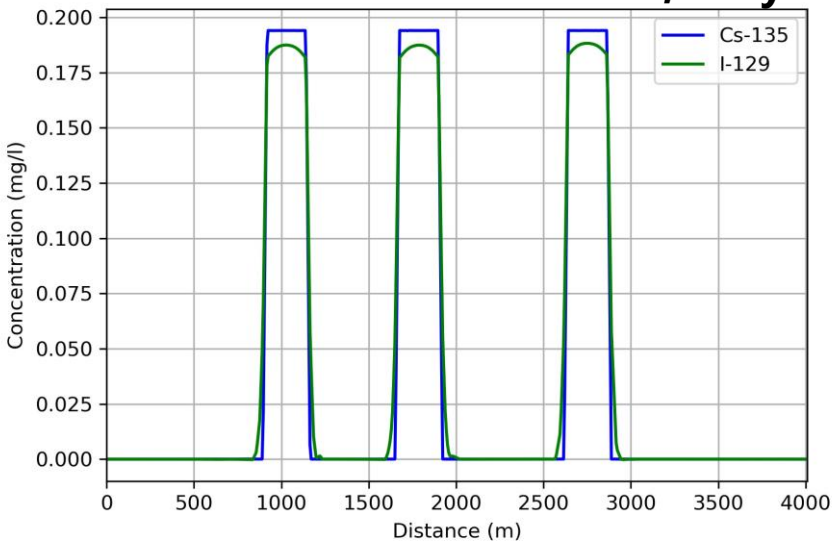
Modelling results: Transport



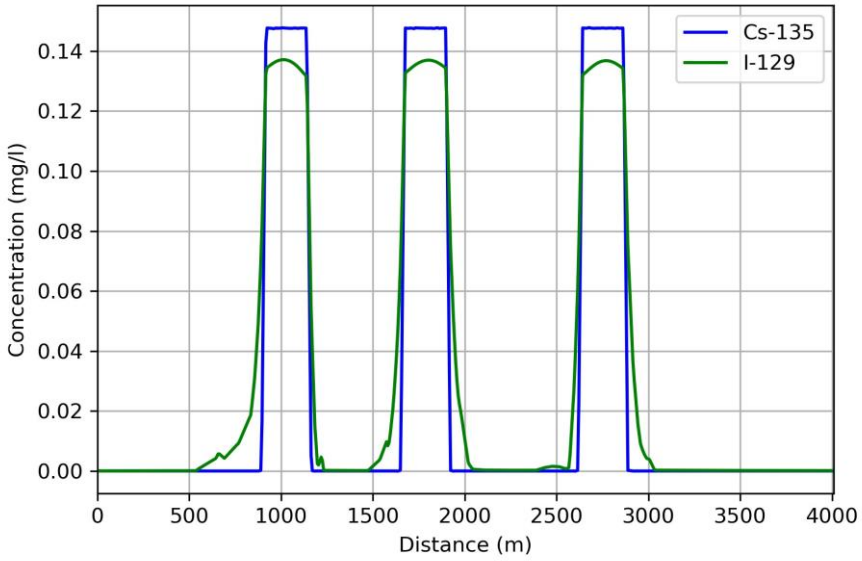
10,000 years



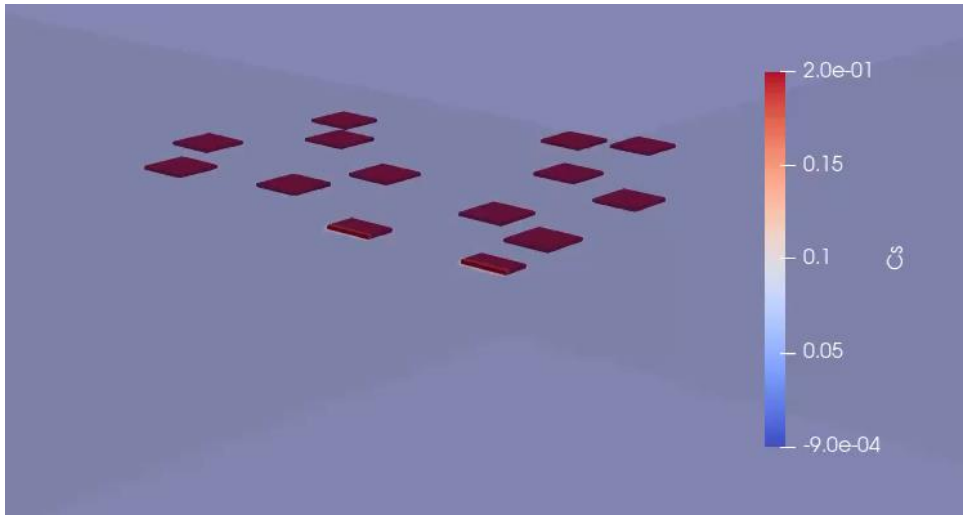
100,000 years

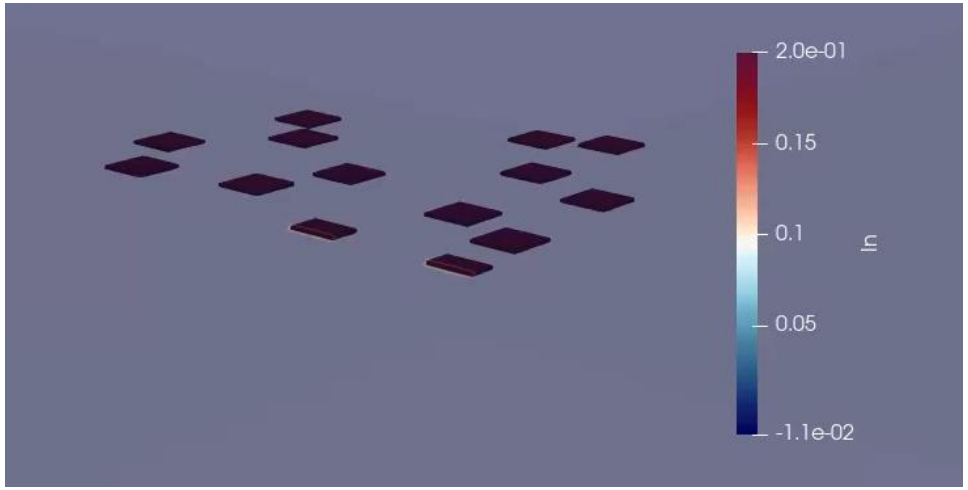


1 Mio years



Migration of CS-135 and I-129 over 1 Mio. Years





- Simulation time: **1 million years**
- Radionuclide release originates from the **repository/deposition zones**
- Migration is governed by the prescribed **groundwater-flow field**
- **Cs-135 and I-129 show different migration behaviour** due to their transport/retention parameter
- sLong-term migration can be evaluated by tracking the **extent and concentration of the plume**

Beyond EPM: Accounting for Fracture Heterogeneity and Uncertainty

Real fracture networks are only partially observable and therefore highly uncertain.
Locations • sizes • orientations • connectivity • apertures / transmissivities cannot be fully characterized from site data

For safety assessment, large scale, long-term and UQ with large realizations are required

Deterministic EPM

k = k_eff

- Fracture effects homogenized
- Preferential pathways smoothed out
- Spatial variability / correlation neglected
- One parameter set → one deterministic prediction

represent heterogeneity

SCM

k = k(x, ω)

- Represents unresolved fracture effects statistically
- Spatial variability + correlation + anisotropy
- Multiple plausible host-rock realizations
- Repository-scale continuum representation

propagate uncertainty

Stochastic analysis

$$k(x,\omega) \rightarrow p(x,\omega), q(x,\omega), C(x,t,\omega)$$

- Propagates random fields through FE equations
- Quantifies uncertainty in flow and transport
- Produces distributions rather than one result
- Supports mean, variance and exceedance metrics

1 deterministic simulation

$t_{sim} \approx 20$ h

→ Classical Monte Carlo

$N_{MC} = 10{,}000$

→ $\underbrace{10{,}000 \times 20\,h}_{\text{brute-force MC}}$ →

Methods to improve the computational efficiency → SFEM

SFEM: Stochastic Finite Element Method







Advanced topic: SFEM

ELSEVIER
Advances in Water Resources
Available online 28 May 2025, 105003
In Press, Journal Pre-proof What's this?

A stochastic modeling framework for radionuclide migration from deep geological repositories considering spatial variability

Zhibao Zheng [a], Xuerui Wang [b] ✉, Judith Flügge [b], Thomas Nagel [c d]

- **Permeability, Porosity** as random fields: extremely high random dimensionality
  - **Flow**

$$S_0\rho\frac{\partial p(t,\mathbf{x},\boldsymbol{\theta})}{\partial t} = -\nabla\cdot\left[\mathbf{K}(\mathbf{x},\boldsymbol{\theta})\cdot\left(\nabla p(t,\mathbf{x},\boldsymbol{\theta})+\mathbf{g}\right)\right]+Q^w$$

  - **Mass transport**

$$\mathbf{D_p} = (\alpha_L-\alpha_T)\frac{\mathbf{v}(\mathbf{x},\boldsymbol{\theta})\otimes\mathbf{v}(\mathbf{x},\boldsymbol{\theta})}{|\mathbf{v}(\mathbf{x},\boldsymbol{\theta})|}+\alpha_T|\mathbf{v}(\mathbf{x},\boldsymbol{\theta})|\mathbf{I}$$

  - **Solution strategy:** SFEM

$$\mathbf{A}(\theta)\,\mathbf{p}(\theta)=\mathbf{B}(\theta) \qquad \mathbf{p}_k(\theta)=\sum_{i=1}^{k}\lambda_i(\theta)\,\mathbf{d}_i=\mathbf{p}_{k-1}(\theta)+\lambda_k(\theta)\,\mathbf{d}_k$$

$$\mathbf{A}(\theta)\,\lambda_k(\theta)\,\mathbf{d}_k=\mathbf{B}_k(\theta)$$

$$\mathbb{E}\left\{\lambda_k^2(\theta)\,\mathbf{A}(\theta)\right\}\mathbf{d}_k=\mathbb{E}\left\{\lambda_k(\theta)\,\mathbf{B}_k(\theta)\right\}$$

$$a_k(\theta)\,\lambda_k(\theta)=b_k(\theta) \text{ with } a_k(\theta)=\mathbf{d}_k^{\mathrm{T}}\mathbf{A}(\theta)\,\mathbf{d}_k\in\mathbb{R}, b_k(\theta)=\mathbf{d}_k^{\mathrm{T}}\mathbf{B}_k(\theta)\in\mathbb{R}$$

Random fields by SCM

⇩

Stochastic finite element discretisation

⇩

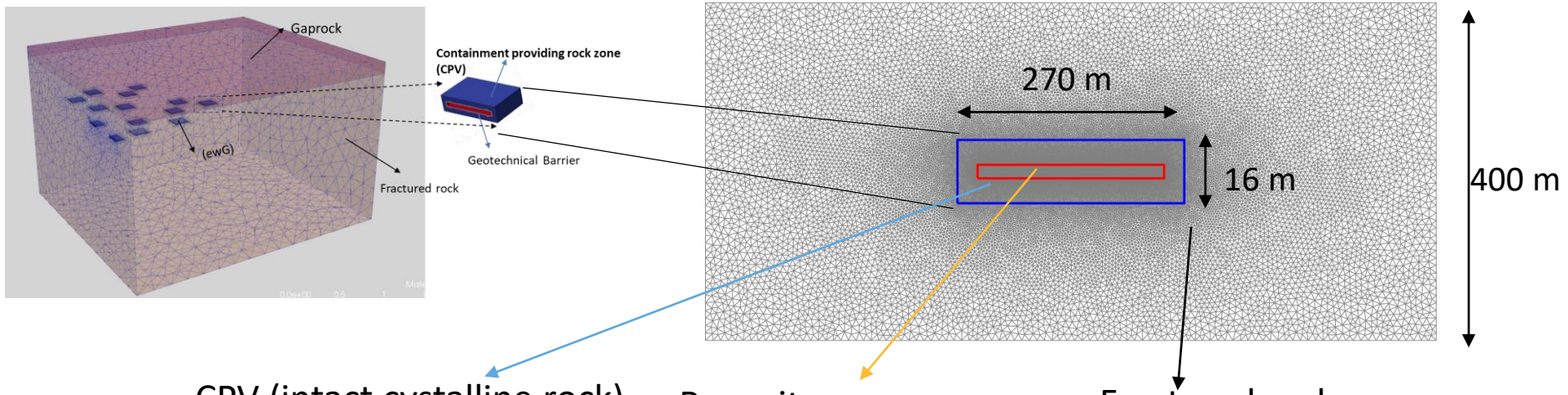
Decoupled approximation of stochastic solutions: $\mathbf{p}_k(\theta)=\sum_{i=1}^{k}\lambda_i(\theta)\mathbf{d}_i$ and others

⇩

Iterative algorithms

Zheng, Z., Wang, X., Flügge, J., & Nagel, T. (2025). A stochastic modeling framework for radionuclide migration from deep geological repositories considering spatial variability. *Advances in Water Resources*, *203*, 105003.

2D Modelling using SFEM



CPV (intact cystalline rock)

$k_c \in [10^{-20}, 10^{-18}]$

Repository area

$k_r \in [10^{-20}, 10^{-18}]$

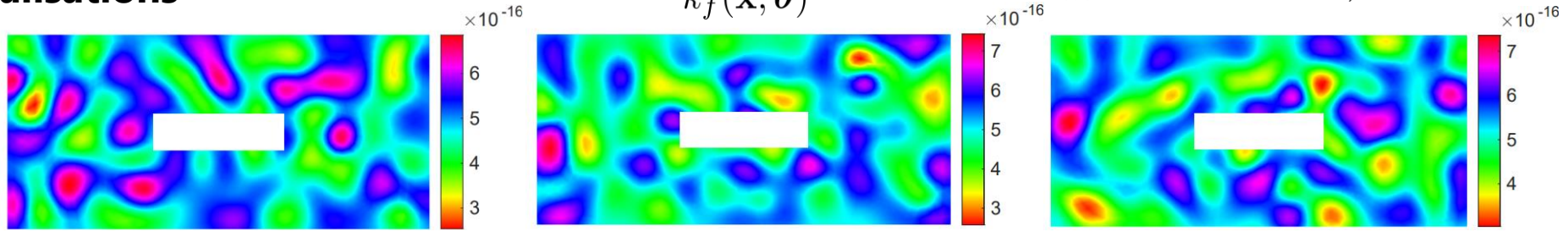
Fractured rock

$k_f(\mathbf{x}, \boldsymbol{\theta})$

$k_m = 5 \times 10^{-16}, \varepsilon = 20\%$

10,000 realisations

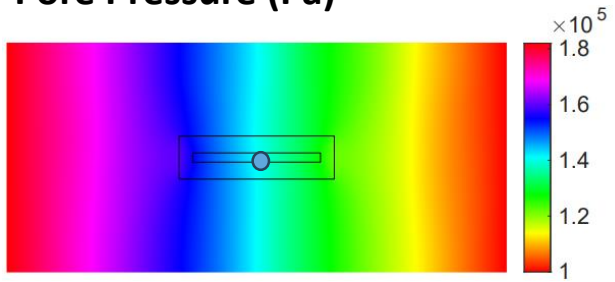
$k_f(\mathbf{x}, \boldsymbol{\theta})$

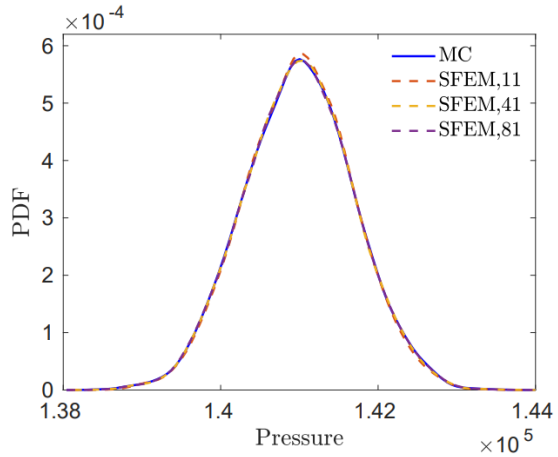


- Transport of I-129 over 1 mio years
- Model verification via Mont-Carlo Simulation (MSC)
- Computational cost: **SFEM vs. MSC → 0.32 vs. 54.44 hours**

Simulation results

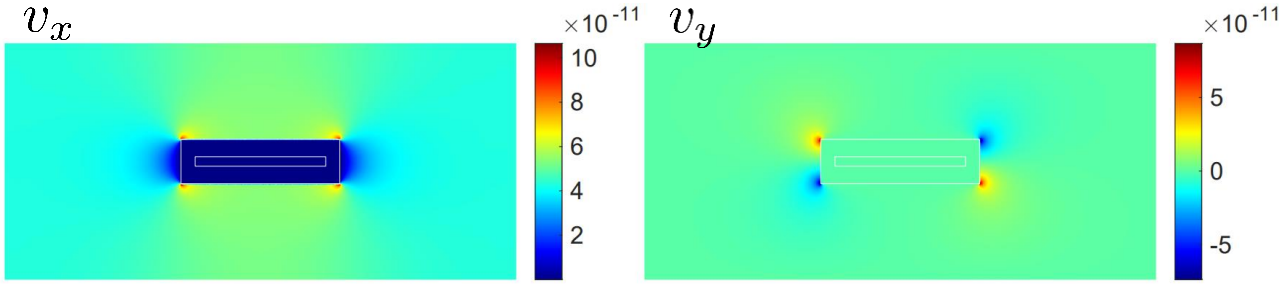
- **Pore Pressure (Pa)**



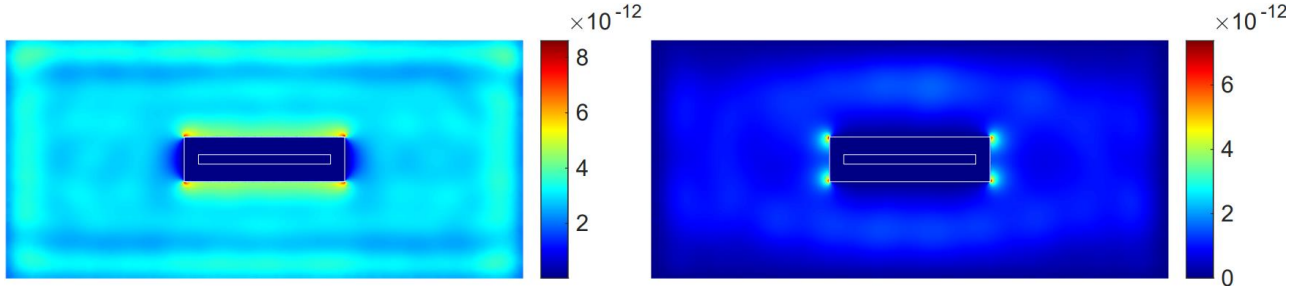


- **Velocity (m/s)**

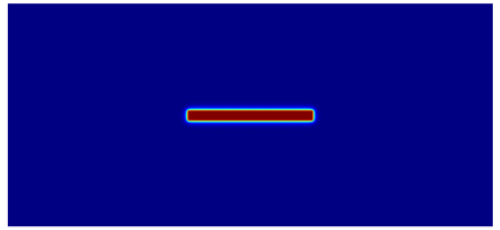
mean



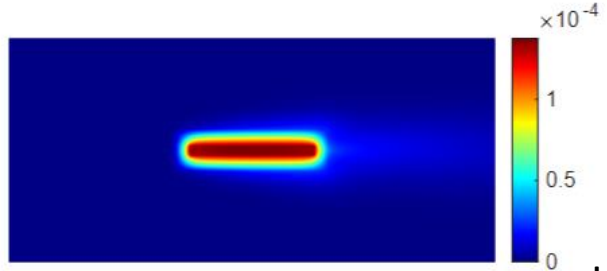
Std



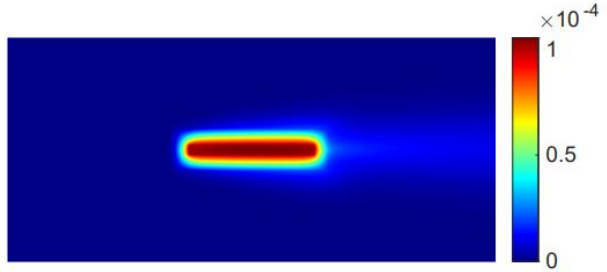
- **Mean Concentration**



t=100 years



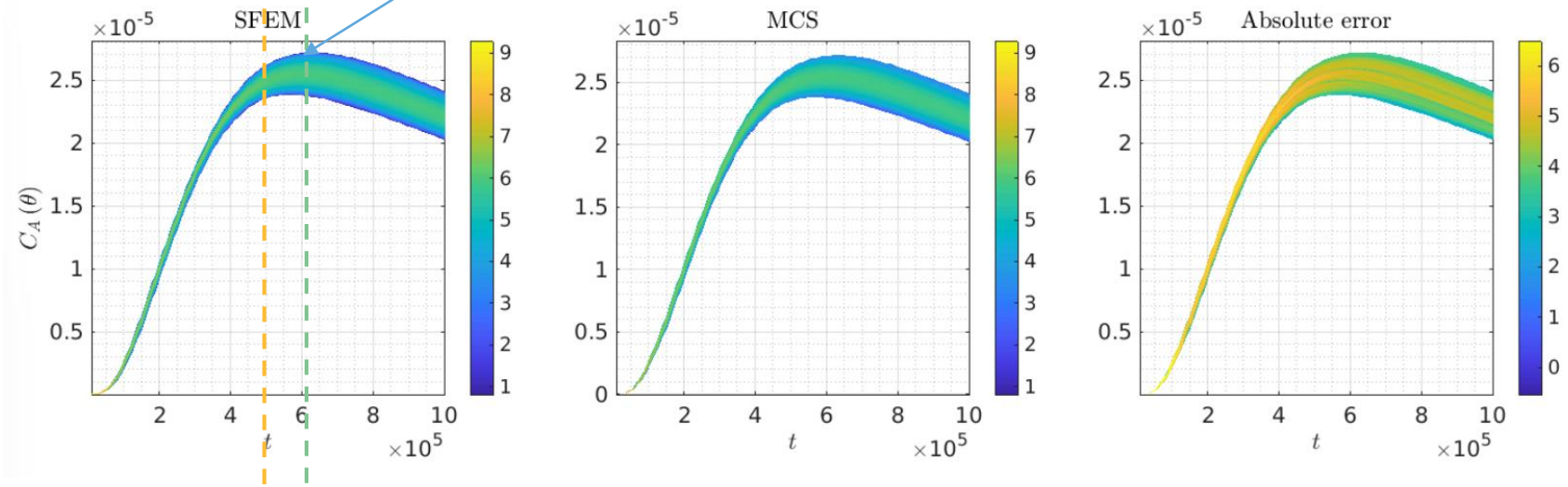
t=0.5 mio years



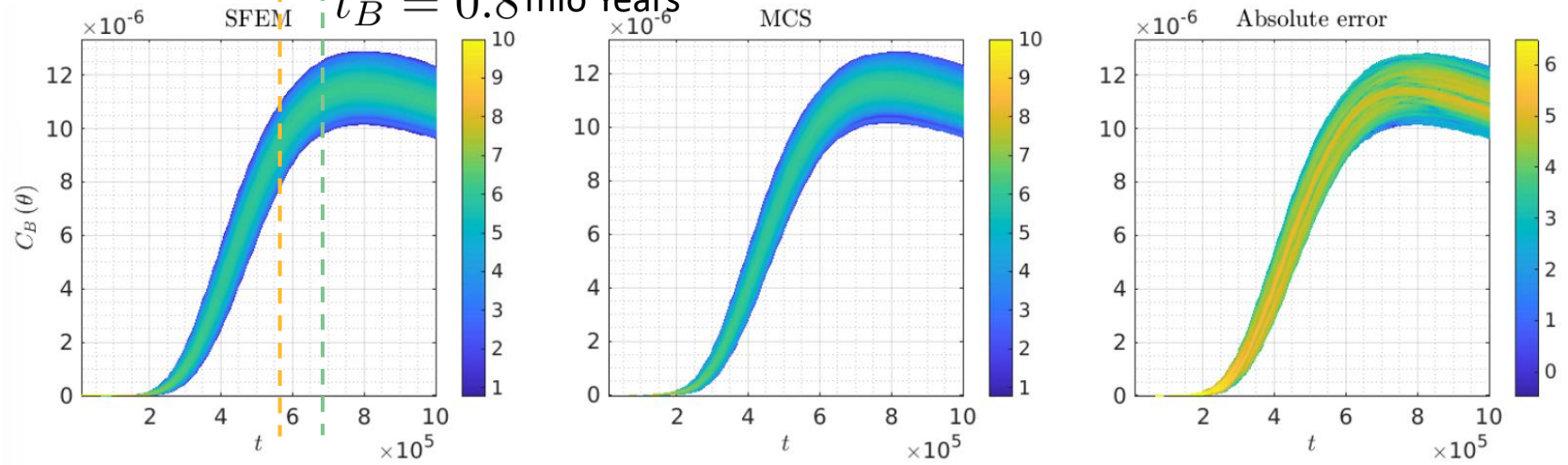
t=1 mio

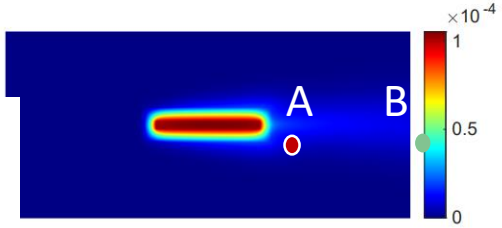
Stochatic analysis

$t_A = 0.6$ mio Years Peak value is relevant for safety analysis!!



$t_B = 0.8$ mio Years





The SFEM will be implemented in OGS !!

Summary & Outlook

From fracture-scale process understanding to repository-scale safety assessment

Key Takeaways

1. **Fracture flow & transport**
   Fractures can form preferential groundwater and radionuclide migration pathways.

2. **Numerical representation**
   Model choice depends on scale and purpose:
   DFN / DFM / LIE / EPM.

3. **OGS implementation**
   LIE provides an explicit lower-dimensional representation for fracture-scale analysis.

4. **Repository application**
   3D EPM enables efficient large-scale flow and radionuclide transport simulations.

Outlook: What Comes Next?

Real fracture system — Incomplete characterization + spatial variability

↓

Efficient UQ — Move beyond brute-force Monte Carlo

↓

Advanced methods — SCM • SFEM • reduced-order / efficient sampling

↓

Safety assessment — Quantify uncertainty in long-term repository evolution

From modelling one possible realization → to quantifying the range and uncertainty of possible repository evolution.

Thank you !!

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